



1. Title of the panel: Quantum CI and AI

2. Short description of the panel

The Panel Session on Quantum Computational Intelligence (CI) and AI at IEEE SSCI 2025 will focus on the emerging intersection of quantum computing and artificial intelligence, with an emphasis on computational intelligence (CI) techniques and their applications. This panel will bring together leading researchers, educators, and industry experts to discuss the future of Quantum CI and AI, bridging theory, applications, and real-world impact.

The panelists include: 1) Giovanni Acampora (University of Naples Federico II, Italy) – A pioneer in Fuzzy Markup Language (FML) and quantum AI, with contributions to evolutionary computation and quantum evolutionary algorithm. 2) Kazi Shah Nawaz Ripon (Oslo Metropolitan University, Norway) – Specializes in bio-inspired AI algorithms, quantum computing, and multi-objective optimization. 3) Samuel Yen-Chi Chen (Wells Fargo, USA) – A quantum machine learning expert with contributions to quantum LSTM and variational quantum circuits. 4) Autilia Vitiello (University of Naples Federico II, Italy) – Focuses on fuzzy logic, computational intelligence, and AI-driven data analysis. 5) Chang-Shing Lee (National University of Tainan, Taiwan) – An expert in human-machine collaborative learning, quantum computational intelligence, and AI education.

Key Discussion Topics: 1) Quantum Evolutionary and Fuzzy Systems for AI Led by Giovanni Acampora; 2) Quantum-Inspired Optimization and AI for Decision-Making Led by Kazi Shah Nawaz Ripon; 3) Quantum Machine Learning (QML) and Its Real-World Applications Led by Samuel Yen-Chi Chen; 4) Computational Intelligence for Quantum AI and Knowledge Representation Led by Autilia Vitiello; 5) Human-Machine Collaborative Learning and Quantum CI in Education Led by Chang-Shing Lee

3. Panelist names and affiliations

Giovanni Acampora, University of Naples Federico II, Italy

Kazi Shah Nawaz Ripon, Oslo Metropolitan University, Norway

Samuel Yen-Chi Chen, Wells Fargo, USA

Autilia Vitiello, University of Naples Federico II, Italy

Chang-Shing Lee, National University of Tainan, Taiwan

4. Date of the panel

Time and Date: 9:00-10:00 AM on March 19, 2025

5. Panelist short bio and their photos

- **Giovanni Acampora, University of Naples Federico II, Italy**



Dr. Giovanni Acampora is a Professor in Artificial Intelligence and Quantum Computing at the Department of Physics "Ettore Pancini," University of Naples Federico II, Italy. He was previously a Reader in Computational Intelligence at Nottingham Trent University (2012–2016) and a Hoofddocent Tenure Track in Process Intelligence at Eindhoven University of Technology (2011–2012). His research focuses on artificial and computational intelligence, fuzzy systems, evolutionary computation, and quantum AI. He developed the Fuzzy Markup Language (FML) and led the IEEE working group that standardized it as IEEE 1855, earning the IEEE-SA Emerging Technology Award. He has been invited to speak at international conferences and universities. Dr. Acampora has served as Secretary and Treasurer of the IEEE CIS Italian Chapter (2010–2012) and chaired the IEEE CIS Standards Committee (2011–2013, 2016–2017). He is Chair of IEEE-SA 1855WG and a Senior IEEE Member. He is Editor-in-Chief of *Springer Quantum Machine Intelligence* and an Associate Editor for *Springer Soft Computing*, *IEEE Transactions on Fuzzy Systems*, and others. He was the General Chair of the FUZZ-IEEE 2017, has organized multiple special sessions, and serves on the scientific board of ICAROS. His awards include the Canada-Italy Innovation Award (2019) for "Quantum Machine Learning" and an IBM Quantum Experience Academic Research Program Award for his work on quantum evolutionary computation. He holds a Full Professor qualification in Italy in Informatics (01/B1).

- **Kazi Shah Nawaz Ripon, Oslo Metropolitan University, Norway**



Dr. Kazi Shah Nawaz Ripon is an Associate Professor in the Department of Computer Science at Oslo Metropolitan University, Norway. His research spans a spectrum of cutting-edge fields, including artificial intelligence, machine learning, quantum computing, combinatorial optimization, bio-inspired algorithms, and deep learning. His work bridges theoretical innovation and practical application by developing AI-driven optimization and decision-making systems that address real-world challenges in healthcare, financial forecasting, logistics, image analysis, industrial optimization, and environmental sustainability. His recent work explores quantum-inspired evolutionary algorithms for complex combinatorial problems and AI-powered diagnostic tools for critical medical conditions. His contributions to bio-inspired computing and deep learning have advanced methods in medical image analysis, fake news detection, and autonomous decision-making. A prolific scholar, he has published extensively in top-tier conferences and journals. Dr. Ripon is actively engaged in applied AI research, collaborating with international experts on projects that encompass quantum AI, autonomous navigation, precision healthcare, and sustainable AI solutions. He also plays a vital role in the academic community by organizing and contributing to AI-focused events and conferences, continually advancing the field of computational intelligence.

- **Samuel Yen-Chi Chen, Wells Fargo, USA**



Dr. Samuel Yen-Chi Chen received his Ph.D. and B.S. degrees in physics, as well as an M.D. degree in medicine, from National Taiwan University, Taipei, Taiwan. He is currently a Lead Research Scientist at Wells Fargo Bank, specializing in quantum machine learning (QML). Previously, he was an Assistant Computational Scientist at the Computational Science Initiative, Brookhaven National Laboratory. Dr. Chen pioneered the use of variational quantum circuits for deep reinforcement learning and is the inventor of quantum LSTM. His research interests focus on developing quantum machine learning algorithms and leveraging classical AI techniques to tackle quantum computing challenges, including quantum error correction and quantum architecture search. He is actively involved in privacy-preserving quantum AI research and is an experienced distributed computing researcher and developer. In 2019, he won First Prize in the Software Competition (Research Category) from Xanadu Quantum Technologies. A seasoned speaker, Dr. Chen has delivered tutorials on quantum machine learning at top-tier conferences. He has presented on quantum neural networks for speech and natural language processing (IJCAI 2021, ICASSP 2022) and quantum tensor networks for signal processing (ICASSP 2024, IJCNN 2024, IEEE QCE 2024). At IEEE ICC 2024, he explored QML applications in 6G communication. Beyond major conferences, Dr. Chen is a frequent speaker at leading research seminars and industry meetups, including the ND-MIT Quantum Computer Systems Lecture Series, Washington DC Quantum Computing Meetup, AQT Seminars at Lawrence Berkeley National Lab, and the Quantum Information Summit at American University of Beirut. He has also presented at ICASSP, IEEE QCE, QTML, NeurIPS, ACM CIKM and IJCNN, covering a wide range of topics in quantum machine learning.

- **Autilia Vitiello, University of Naples Federico II, Italy**



Dr. Autilia Vitiello is an Assistant Professor at the Department of Physics “Ettore Pancini,” University of Naples Federico II, since December 2018. She was a Research Fellow at the University of Salerno (2013–2018) and a visiting student at Eindhoven University of Technology in 2012. She holds a Ph.D. in Computer Science (2013) and an M.Sc. **Laurea cum Laude** (2009) from the University of Salerno, focusing on memetic algorithms for ontology alignment and time-sensitive fuzzy agents. Her academic background includes international summer schools in software engineering and computational intelligence. Dr. Vitiello has taught Informatics, Cybernetics, and Databases at the University of Naples Federico II and the University of Salerno. She has held leadership roles in the IEEE Computational Intelligence Society (CIS), serving as Chair of the IEEE CIS Standards Committee and the Task Force on Datasets for Computational Intelligence Applications. She is also a member of the IEEE Standard Association Working Group for Fuzzy Markup Language and serves on the editorial boards of *Scientific Programming* and *Quantum Machine Intelligence*. Her research in computational intelligence has earned recognition, including the Best Paper Award at the UK Workshop on Computational Intelligence (2012). She is a member of the Accademia Italiana di Scienze Forensi (ACISF) and the IEEE CIS (since 2010) and received a national qualification as an Associate Professor in 2019.

- **Chang-Shing Lee, National University of Tainan, Taiwan**



Dr. Chang-Shing Lee (IEEE Senior Member) is a Professor at the Department of Computer Science and Information Engineering, National University of Tainan, Taiwan. He earned his Ph.D. from National Cheng Kung University in 1998. He has served in various leadership roles, including Director of the Computer Center (2006–2011) and Dean of the Research and Development Office (2011–2015). His contributions to AI education and research have been recognized with multiple awards, including the IEEE EAB Meritorious Achievement Award in Pre-University Education (2022) and the MOST FutureTech Breakthrough Award (2017). Prof. Lee has held several key IEEE roles, including Chair of the IEEE CIS Outreach Subcommittee (2025), IEEE CIS Education Portal Subcommittee (2023–2024), Chair of the IEEE CIS High School Outreach Subcommittee (2021–2022), and Chair of the IEEE CIS Tainan Chapter (2015–2017). He has also served as an associate editor for leading journals, including IEEE Transactions on Fuzzy Systems, IEEE Transactions on Games, and Quantum Machine Intelligence. He has pioneered research in human-machine collaborative learning, computational intelligence, and knowledge ontology, publishing extensively with significant academic impact. He played a key role in standardizing IEEE 1855 (Fuzzy Markup Language) and received the IEEE Outstanding Contribution Certificate for his efforts in 2016. Prof. Lee has been actively involved in organizing major IEEE conferences and competitions, including FUZZ-IEEE, IEEE WCCI, IEEE CEC, and IEEE SMC. Since 2017, he has been dedicated to promoting Quantum Computational Intelligence (QCI) and AI education through international workshops and competitions, inspiring young students worldwide.



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@ IEEE SSCI 2025**

• Venue: Cosmos 1&2, Clarion Hotel & Congress Centre, Trondheim, Norway
• Time and Date: 9:00–10:00 AM on March 19, 2025



Webpage



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